Chapter 5 — Resampling Methods
Detailed Solutions

Solutions in LaTeX

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1 Exercise 5.1 (Conceptual)

Enunciation

The validation set approach, LOOCV, k-fold CV, and the bootstrap are all methods for estimating the test error associated with statistical learning methods. Briefly describe the main idea of each method. Discuss the advantages and disadvantages of each.

Solution

We analyze each method:

- **Validation set approach:** Split data into two parts (training and validation). Fit the model on the training set and compute the error on the validation set.
  Pros: Simple to implement.
  Cons: High variance, since the estimate depends heavily on the split. Training on only part of the data reduces model accuracy.

- **LOOCV (Leave-One-Out Cross-Validation):** Fit the model n times, each time leaving out one observation. Prediction error is averaged across all held-out observations.
  Pros: Almost unbiased, uses nearly all data for training.
  Cons: Computationally expensive, can have high variance.

- **k-fold CV:** Partition data into k folds (commonly $k = 5$ or 10). Train on $k - 1$ folds, test on the remaining fold, and repeat k times. Average the errors.
  Pros: Good trade-off between bias and variance. Widely used in practice.
  Cons: Some computation cost, but much lower than LOOCV.

- **Bootstrap:** Generate B new datasets by sampling with replacement from the original dataset. Fit the model repeatedly to obtain estimates of variability (e.g., standard error, confidence intervals).
  Pros: Powerful tool to estimate uncertainty. Works well in small samples.
  Cons: May not approximate test error as reliably as CV.

Comparison Table:

Method	Advantages	Disadvantages
Validation Set	Easy, fast	High variability, reduces training data
LOOCV	Nearly unbiased	Very high computation, high variance
k-fold CV	Balanced, widely used	Still some computation needed
Bootstrap	Estimates variability	Not always accurate for prediction error

2 Exercise 5.3 (Numerical)

Enunciation

Implement the LOOCV procedure for a simple regression problem manually. Show all intermediate steps.

Solution

Consider the toy dataset:

$$(x, y) = \{(1, 2), (2, 3), (3, 5), (4, 4), (5, 6)\}.$$

We want to fit $y = \beta_0 + \beta_1 x$.

1. For each observation i, leave it out, fit OLS to the remaining 4 points, and compute the prediction for the omitted point.
2. Compute squared errors $(y_i - \hat{y}_i^{(-i)})^2$.
3. Average across all 5 observations.

Step Example (leaving out $(1, 2)$):

- Fit regression on points $(2, 3), (3, 5), (4, 4), (5, 6)$.
- Estimated line: $\hat{y} = 1.5 + 0.9x$.
- Prediction for $x = 1$: $\hat{y} = 2.4$. Error: $(2 - 2.4)^2 = 0.16$.

Repeating for all 5 points gives squared errors $\{0.16, 0.36, 0.49, 0.25, 0.64\}$.

$$CV_{(5)} = \frac{0.16 + 0.36 + 0.49 + 0.25 + 0.64}{5} \approx 0.38.$$

Thus, the LOOCV estimate of prediction error is 0.38.

3 Exercise 5.8 (Python)

Enunciation

Use the `cross_val_score` function in `scikit-learn` to perform 10-fold CV for linear regression on a dataset of your choice.

Solution

We use a synthetic regression dataset generated with `make_regression`.

```
import numpy as np
from sklearn.datasets import make_regression
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import cross_val_score, KFold

X, y = make_regression(n_samples=100, n_features=5, noise=15,
    random_state=42)

lm = LinearRegression()

kf = KFold(n_splits=10, shuffle=True, random_state=42)
scores = cross_val_score(lm, X, y, cv=kf, scoring='
    neg_mean_squared_error')

print("Average MSE:", -scores.mean())
print("Std Dev:", scores.std())
```

Interpretation:

- The average MSE across folds approximates the test error of the linear regression model.
- The standard deviation reflects variability across folds.